

Ajay Venkatesh

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Summary

Software Engineer with 4+ years of experience building scalable backend systems, payment platforms, financial data pipelines, and cloud-native solutions across fintech, insurance, enterprise consulting, and fulfillment domains. Proficient in Java, Python, Node.js, RESTful APIs, AWS, Azure, Kafka, Docker, and Kubernetes. Experienced in transaction processing, event-driven architectures, real-time data streaming, secure APIs, RBAC, compliance-driven workflows, and AI-powered solutions using LLMs, LangChain, and RAG.

Skills

- Programming Languages:** Java, Python, C++, C#, TypeScript, JavaScript, SQL
- Web Development:** Node.js, Express.js, React.js, HTML, CSS, RESTful APIs, GraphQL APIs
- Databases and Caching:** PostgreSQL, MySQL, DynamoDB, Redis, Database Design, Query Optimization
- Cloud:** AWS (CDK, Lambda, API Gateway, EC2, S3, DynamoDB, IAM), Azure (OpenAI, Functions, Storage)
- Professional Skills:** Cross-Functional Collaboration, Stakeholder Management, Requirement Analysis, Mentoring, Team Collaboration
- DevOps:** Docker, Kubernetes, CI/CD, GitHub Actions, IaC, Serverless Framework
- AI and Generative AI:** LLMs, RAG, OpenAI APIs, LangChain, NLP, OCR, Prompt Engineering
- Architecture & Integration:** System Design, API Integration, Microservices, Distributed Systems, Event-Driven Architecture, Low-Level Design
- Software Engineering:** Design Patterns, SDLC, Agile Scrum, Code Reviews, Technical Documentation, Mentoring
- Security and Quality:** OAuth, SSO, RBAC, KMS, SSM, Secure Coding, Unit Testing, Integration Testing, Mockito, Debugging

Experience

CampusMesh

Software Engineer

Jan 2026 – Present

New York, US

- Built a **Node.js** serverless **Microservices** backend with layered **handlers, services, and models**, using **REST APIs, AWS Chime, DynamoDB**, and **MySQL** to power real-time **chat, video calls, and document sharing** for the study group feature, engaging **10K+** students across **100+** study groups.
- Built a personalized **recommendation engine** using **collaborative filtering** and behavioral analytics to surface relevant posts, users, study groups, events, and content for **10K+** users, increasing engagement by **28%** and retention by **50%**.
- Standardized serverless deployment pipelines across **4 environments** using **AWS Serverless Framework, SSM Parameter Store**, and **KMS** with reusable configuration and least-privilege **IAM** to enforce consistent release practices, cutting deployment setup time by **90%**.
- Engineered **CampusIQ**, an agentic AI chatbot powered by **RAG** (LangChain, FAISS, AWS Bedrock) with tool-calling and multi-step reasoning to resolve student queries on events, study groups, and course content, improving contextual accuracy by **30%** and reducing manual support overhead.

Amazon

Software Development Engineer Intern

May 2025 – Aug 2025

Seattle, US

- Designed a **UI console for Amazon FC** with **AuthN and AuthZ** across multiple roles using **OAuth, Midway SSO, and HELIS** to centralize large-scale **video and image evidence** review with secure role-based access; presented the system design to stakeholders for approval.
- Provisioned **Infrastructure as Code** using **AWS CDK** and **CloudFormation** stacks to enable repeatable multi-region deployments; configured **multi-stage CI/CD pipelines** across dev, staging, and production, reducing provisioning time by **70%**.
- Built a **React 18** frontend with **lazy loading, client-side caching**, and **URL-synchronized filter state** via query parameters to enable **deep linking** and shareable filter context across FC teams, reducing page load time by **40%** on media-heavy evidence views.
- Engineered serverless **Lambda Java** backend on **REST Architecture** using **Coral, Smithy, Dagger**, and **CBOR serialization** with code design patterns to handle data fetching, reducing execution time by **35%** and ensured reliability through unit and integration testing with **Mockito**.

New York University (Novel AI Technologies, Inc.)

Software Developer

Aug 2024 – Present

New York, US

- Architected **Firebeats**, an NSF-funded (\$100K) health platform using **React, Firebase, GCP**, and **HealthKit**, to collect and visualize real-time wearable device data alongside food logs into personalized health dashboards, across **iOS and Android**.
- Engineered **ETL pipelines** using **Python** and **SQL queries** to migrate and transform **250K+** records from **Firebase** into **PostgreSQL**, with scheduled jobs, triggers, transaction locking, and race-condition guards; built **AWS QuickSight** dashboards for health analytics.
- Built a **financial CRM platform** hosted on **AWS EC2** with **Docker**, embedding **QuickSight** dashboards and enforcing **RBAC** via **IAM** for company-scoped data access; integrated **Stripe** payment processing for policy purchases, generating **\$50K+** in revenue across **6** clients.
- Developed a **broker portal** using **Node.js** and **Express.js** on the same EC2 deployment, handling financial form submissions routed via **Kafka** to carriers for underwriting and claims management; integrated **AWS SES** for automated notifications and **WebSocket** real-time chat.

PricewaterhouseCoopers (PwC)

Senior Software Engineer

Aug 2021 – Aug 2024

Bengaluru, India

- Architected **RESTful APIs** and backend services using **Spring Boot, Node.js**, and **Express.js**, to streamline customer onboarding and workflow automation, reducing manual processing effort for business operations teams.
- Implemented cloud-native solutions on **Azure** utilizing **VMs, Blob Storage, Azure Functions**, and **SQL Database**, containerized with **Docker** and orchestrated via **Kubernetes**, to improve application scalability and operational flexibility.
- Deployed an **NLP-based OCR** pipeline using **Azure Document Intelligence**, to automate structured data extraction from enterprise health records, cutting manual processing time by **75%**.
- Enhanced **data ingestion** and processing workflows using **Python, SQL**, and **PostgreSQL** with **parallel processing**, to support reporting and business intelligence requirements, cutting execution time from **5 hours to 1 hour** and improving data availability for stakeholders.
- Partnered with cross-functional teams in **Agile** environments to modernize legacy applications into **microservice-based architectures**, improving maintainability, scalability, and long-term operational efficiency across enterprise systems.

Project

Advanced Expense Tracker | Java, JavaFX, JDBC, SQLite, JFreeChart, OOP, MVC, DAO

- Architected a modular **expense management** desktop application using **Java, JavaFX, JDBC**, and **SQLite** with **MVC** and **DAO** patterns to support secure multi-user login, expense tracking, and group cost splitting across shared workflows.
- Implemented **concurrent transaction handling** and real-time cost splitting using **Java multithreading** and synchronized blocks to ensure thread-safe database updates across shared expense workflows, eliminating race conditions under concurrent multi-user load.

Education

New York University — M.S. in Computer Engineering | GPA: 3.9/4.0

Aug 2024 – May 2026

Relevant Coursework: System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Visvesvaraya Technological University — B.E. in Electronics and Communication | GPA: 8.9/10

Jun 2017 – Aug 2021